

Project Reflection

Project 002 – Sentiment Analysis

1. Introduction

This project provided practical experience in sentiment annotation by requiring the manual classification of customer feedback across a variety of real-world scenarios. While assigning sentiment labels initially appeared straightforward, the project demonstrated that consistent annotation depends on careful interpretation of customer intent, context, and expectation rather than isolated words or phrases.

2. What I Learned

Accurate annotation requires understanding the overall message being communicated. And not just associating labels to specific words. Throughout the project, I learned to distinguish between factual status updates and genuine customer opinions, identify dominant sentiment in complex messages, and recognize when multiple viewpoints should be represented using the Mixed label. I also learned to internalize the guidelines and always fall back to it when I'm met with ambiguous cases. Generally, I developed a greater appreciation for consistency, particularly when reviewing similar examples across different contexts.

3. Challenges Encountered

Several aspects of the project proved challenging. Messages containing both positive and negative opinions required careful evaluation to determine whether both sentiments contributed meaningfully to the overall message or whether one clearly dominated.

Another challenge involved distinguishing successful outcomes from positive sentiment. Many customer messages described completed actions or restored functionality without expressing satisfaction, making it necessary to rely on context rather than assumptions.

Product reviews also introduced difficult decisions where positive cosmetic features had to be weighed against failures in core functionality or unmet customer expectations.

4. Quality Assurance Experience

The QA process became one of the most valuable parts of this project. Rather than simply correcting labels, reviewer feedback encouraged careful discussion of ambiguous examples and promoted consistent application of the annotation guidelines.

Several adjudication discussions resulted in refinements to the project's annotation principles, including:

- distinguishing status updates from sentiment,
- evaluating service recovery,
- identifying expectation violations,
- assessing core functionality versus secondary product attributes, and
- recognizing when restored functionality should not automatically be interpreted as positive sentiment.

These discussions strengthened both the annotation guidelines and the final dataset.

5. Skills Developed

Completing this project improved my ability to:

- apply consistent annotation guidelines,
- analyze ambiguous customer feedback,
- justify annotation decisions using textual evidence,
- perform structured quality assurance reviews,
- identify annotation inconsistencies,
- participate in reviewer calibration, and
- document annotation workflows professionally.

These skills are directly applicable to AI data annotation, NLP dataset development, and LLM evaluation projects.

6. Areas for Improvement

Although the final annotation quality improved significantly, the project highlighted opportunities for continued development. Future projects can further strengthen consistency by expanding

annotation guidelines before annotation begins, increasing the diversity of edge cases, and incorporating additional reviewers to measure inter-annotator agreement. These improvements would better reflect large-scale commercial annotation workflows.

Overall, this project reinforced the importance of consistency, decision-making, and continuous guideline refinement when producing reliable training data for AI systems. Completing this project strengthened both my technical annotation skills and my understanding of professional quality assurance practices.